

MOHAMMED RAKIB

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Portfolio

Summary

Computer Science PhD candidate with experience across machine learning engineering, data science, applied research, and backend software development. Built production-oriented ML and data systems using Python, PyTorch, Hugging Face, FastAPI, Docker, Kubernetes, AWS, and distributed processing, with applications in failure prediction, neural machine translation, NLP, computer vision, and multimodal learning. Published first-author research at top AI/ML venues, including ICCV, KDD, and ICLR, with demonstrated impact in building scalable, measurable, and deployable AI solutions.

Education

Doctor of Philosophy, Computer Science

Oklahoma State University (OSU)

• GPA: 3.88

May 2025 - Present

Stillwater, OK, USA

Master of Science, Computer Science

Oklahoma State University (OSU)

• GPA: 3.82

• Coursework: Big Data Analytics, Cloud Computing & Distributed Systems, Data Structure & Algorithms II, Database Systems

Aug 2023 - May 2025

Stillwater, OK, USA

Bachelor of Science, Computer Science and Engineering

North South University (NSU)

• GPA: 3.96

Sep 2017 - Sep 2021

Dhaka, Bangladesh

Technical Skills

- **AI & Data Science:** PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, OpenCV, Pandas, Matplotlib, SQL, Natural Language Processing, Computer Vision, Seaborn
- **Big Data Technologies:** Apache Spark, Hadoop
- **DevOps & Containerization:** Docker, Kubernetes, Rest APIs
- **Cloud Computing:** AWS (EC2, S3)
- **Programming Languages:** Python, Java, C++, C, SQL, Shell Scripting
- **Databases:** MySQL, MongoDB, Redis, PostgreSQL, NoSQL
- **Web Development:** Django, FastAPI, Flask, Nginx, JavaScript, HTML, CSS

Work Experience

eBay

Software Engineer Intern

- Built a failure prediction dataset by scraping, cleaning, and curating system logs and metrics, then trained ML models to predict failures 30-60 mins early, targeting >80% precision and >70% recall.
- Created a preemptive migration trigger pipeline that detects likely pod, node, server, or cluster failures and starts volume migration before users are affected, cutting visible downtime from 30s-5mins to under 10s.

May 2026 - Aug 2026

San Jose, CA, USA

Neovo Tech Ltd.

Data Scientist & Machine Learning Engineer

- Developed a distributed web crawling system for fintech client Eucaps using Python, Celery, and RabbitMQ to scrape and ingest financial news, containerized with Docker and orchestrated on Kubernetes, which increased daily news coverage by delivering timely data to downstream analytics
- Engineered a neural machine translation pipeline that processed scraped news with a fine-tuned Hugging Face model, deployed in Docker containers and stored translations in AWS S3, enabling multilingual access to financial updates for the client
- Co-developed the backend of the Padel Mates booking platform using Python, FastAPI, and Docker, creating over 30 RESTful API endpoints for authentication, scheduling, and Stripe payments, which streamlined the booking process with reliable user transactions

Jun 2022 - Jul 2023

Dhaka, Bangladesh

Complex Systems Lab, Oklahoma State University

Graduate Research Assistant

- Spearheaded research in multimodal learning, developing the G2D framework with PyTorch to mitigate modality imbalance, which resulted in a first-author publication at ICCV 2025
- Engineered a multimodal fusion model (MIS-ME) for soil moisture estimation, reducing prediction error (MAPE) by over 3% and resulting in a first-author publication at DSAA 2024.

Aug 2023 - Present

Stillwater, OK, USA

Department of Computer Science, Oklahoma State University

Graduate Teaching Assistant

- Serve as a TA for Design & Implementation of Operating Systems I, Intro to Object-Oriented Programming, Database Systems, and Discrete Mathematics, leading weekly labs, grading assignments, and reviewing code with C++, Java, MySQL, and shell scripts which helped students improve their understanding and pass rates

Aug 2023 - Present

Stillwater, OK, USA

- Developed and deployed deep learning models for NLP (QA, NER) and Computer Vision (ASR, OCR) tasks, resulting in publications at ICPR 2022 and ICLR 2023 (Tiny Papers).
- Researched model-compression methods and created an iterative pruning algorithm (COLT) with PyTorch, maintaining accuracy comparable to unpruned models, which led to a publication in IEEE Transactions on AI

Selected Projects**Gradient-Guided Distillation for Multimodal Learning (G²D)** <https://github.com/rAISON-Lab/G2D> **Aug 2024 - Mar 2025**

- Authored and open-sourced G²D, a novel knowledge distillation framework that mitigates modality imbalance by introducing a custom loss function and a Sequential Modality Prioritization (SMP) technique.
- Outperformed 10+ state-of-the-art methods on 6 benchmark datasets, notably improving classification accuracy by over 18% on the CREMA-D dataset with only a 0.5% memory overhead compared to standard baselines.

PhyGHT: Physics-Guided HyperGraph Transformer for Signal Purification at the HL-LHC **Oct 2025 - Mar 2026**<https://github.com/rAISON-Lab/PhyGHT>

- Developed PhyGHT, a physics-guided hypergraph transformer for HL-LHC pileup mitigation, achieving $R^2=0.932$ for energy and $R^2=0.836$ for mass correction under extreme pileup.
- Reduced per-event inference latency to 40.4 ms, delivering $\sim 1.9\times$ faster inference than PUMiNet and $\sim 8.7\times$ faster inference than ParticleNet while maintaining the best reconstruction accuracy.

Distributed k-NN Classifier with Hadoop MapReduce **Jan 2024 - Mar 2024**

- Engineered a distributed k-NN classifier from scratch in Python for the Hadoop MapReduce framework, designing a memory-efficient Reducer using a priority queue to find top-k neighbors.
- Deployed and benchmarked the system across single-node and multi-node Hadoop clusters, analyzing the trade-offs between runtime performance and classification accuracy for various 'k' values.

Bangla-Wave: Improving Bangla Automatic Speech Recognition Utilizing N-gram Language Models **Dec 2022 - Jul 2023**<https://arxiv.org/abs/2209.12650>

- Finetuned a pretrained wav2vec2 model on the Bengali Common Voice dataset (399 hours) to develop a state-of-the-art Bangla ASR model, achieving a WER of 4.66% and a CER of 1.54%.
- Enhanced transcription accuracy by integrating an n-gram language model as a post-processor, optimizing hyperparameters, and implementing preprocessing techniques for improved generalization.

An Open Source Contractual Language Understanding Application Using Machine Learning **Aug 2022 - Dec 2022**<https://github.com/afra-tech/defactolaw>

- Co-lead a team of 4 to develop an end-to-end legal contract review system that processes digital and scanned contracts, using transformer-based models to highlight key clauses and provide contextual explanations for non-legal users.
- Fine-tuned and optimized RoBERTa-base, improving AUPR by 4%, and deployed the model in a resource-constrained environment; made it publicly available on Hugging Face, where it receives 20k downloads per month.

Selected Publications

- M. Rakib, et al..PhyGHT: Physics-Guided HyperGraph Transformer for Signal Purification at the HL-LHC..*KDD 2026*.
- M. Rakib, et al..G²D: Boosting Multimodal Learning with Gradient-Guided Distillation..*ICCV 2025*.
- M.I. Hossain, M. Rakib, et al..COLT: Cyclic Overlapping Lottery Tickets for Faster Pruning of Convolutional Neural Networks..*IEEE Transactions on AI (TAI)*, 2025.
- M. Rakib, et al..MIS-ME: A Multi-modal Framework for Soil Moisture Estimation..*DSAA 2024*.
- S. Mollah, M. Rakib, et al..Automated Mapping of Healthcare Concepts to a Standardized Healthcare Taxonomy..*ICLR 2023*
- M.I. Hossain, M. Rakib, et al..LILA-BOTI: Leveraging Isolated Letter Accumulations by Ordering Teacher Insights.....*ICPR 2022*.

Awards and Honors

- **1st Place** - eBay VibeCoding Hackathon, 2026.
- **Research Credits Award** worth \$1000 by Google Cloud, 2026
- **Student Travel Award** worth \$400 by OSU Graduate College, 2025
- **Runner Up**: Bengali ASR Competition, DL Sprint - BUET CSE Fest, 2022.
- **Summa Cum Laude** Distinction at NSU for achieving a GPA of 3.96, 2021
- **Academic Excellence Award** at NSU – 75% scholarship for CSE Bachelor's Degree, 2017

Academic Service

- Served as a **Reviewer** for CS Venues: Neural Networks, BMVC'26, CVPR'25, ICCV'25, & IJCNN'24.
- Led a **DataBytes Workshop** on Multimodal Learning for graduate students and faculty at OSU.
- Instructed hands-on **PyTorch & Deep Learning Workshops** at NSU to equip students with practical neural network skills.